

WEEK 2 MODULE

1. Fraud Transaction Detection (Linear Search)

```
import java.util.Scanner;
```

```
public class LinearSearchExample {  
    public static void main (String[] args) {
```

```
        Scanner sc = new  
Scanner (System.in);
```

```
        int n = sc.nextInt();
```

```
        int[] transactions = new int[n];
```

```
        for (int i=0; i<n; i++) {  
            transactions[i] = sc.nextInt();  
        }
```

```
        int target = sc.nextInt();
```

```
        boolean found = false;
```

```
        for (int i=0; i<n; i++) {  
            if (transactions[i] == target) {  
                found = true;  
                break;
```

```
            }
```

```
        }
```

```
if (found) {
```

```
    System.out.println("Fraud Transaction Found");
```

```
} else {
```

```
    System.out.println("Fraud Transaction Not  
Found");
```

```
}
```

```
    sc.close();
```

```
}
```

```
}
```

```
import java.util.Scanner;
```

```
public class LinearSearchExample {
```

```
    public static void main(String[] args) {
```

```
        // TODO Auto-generated method stub
```

```
        Scanner sc = new Scanner(System.in);
```

```
        // Read size of array
```

```
        int n = sc.nextInt();
```

```
        // Store transaction IDs
```

```
        int[] transactions = new int[n];
```

```
        for (int i = 0; i < n; i++) {
```

```
            transactions[i] = sc.nextInt();
```

```
        }
```

```
        // Suspicious transaction ID to search
```

```
        int target = sc.nextInt();
```

```
        boolean found = false;
```

```
        // Linear Search
```

```
        for (int i = 0; i < n; i++) {
```

```
            if (transactions[i] == target) {
```

```
                found = true;
```

```
                break;
```

```
            }
```

```
        }
```

```
        // Output
```

```
        if (found) {
```

```
            System.out.println("Fraud Transaction Found");
```

```
        } else {
```

```
            System.out.println("Transaction Not Found");
```

```
        }
```

```
        sc.close();
```

```
    }
```

```
}
```

Problems Javadoc Declaration Console X Progress

<terminated> LinearSearchExample [Java Application] /Applications/Eclipse.app/Contents/Eclipse/plugins/org.eclipse.justi.openjdk.hotspot.jre.full.macosx.aarch64_21.0.9.v20251105-0741/jre/bin/java (Jun

1201 1305 1450 1408 1502

1450

Fraud Transaction Found

2. Server Log Search (Binary Search)

```
import java.util.Scanner;

public class BinarySearchExample {
    public static void main (String [ ] args) {
        Scanner sc = new
Scanner (System.in);

        int n = sc.nextInt();

        int [ ] logs = new int [n];

        for (int i=0; i<n; i++) {
            logs [i] = sc.nextInt();
        }

        int target = sc.nextInt();

        int low = 0;
        int high = n-1;
        boolean found = false;

        while (low <= high) {
            int mid = (low + high) / 2;

            if (logs [mid] == target) {
                found = true;
                break;
            } else if (target < logs [mid]) {
                high = mid - 1;
            } else {
                low = mid + 1;
            }
        }
    }
}
```

```
if (found) {
```

```
    System.out.println("Log Found");
```

```
} else {
```

```
    System.out.println("Log Not Found");
```

```
}
```

```
sc.close();
```

```
}
```

```
}
```



```

3 import java.util.Scanner;
4
5 public class BinarySearchExample {
6
7     public static void main(String[] args) {
8         // TODO Auto-generated method stub
9         Scanner sc = new Scanner(System.in);
10
11         int n = sc.nextInt();
12
13         int[] logs = new int[n];
14
15         for (int i = 0; i < n; i++) {
16             logs[i] = sc.nextInt();
17         }
18
19         int target = sc.nextInt();
20
21         int low = 0;
22         int high = n - 1;
23         boolean found = false;
24
25         while (low <= high) {
26             int mid = (low + high) / 2;
27
28             if (logs[mid] == target) {
29                 found = true;
30                 break;
31             } else if (target < logs[mid]) {
32                 high = mid - 1;
33             } else {
34                 low = mid + 1;
35             }
36
37             if (found) {
38                 System.out.println("Log Found");
39             } else {
40                 System.out.println("Log Not Found");
41             }
42
43             sc.close();
44         }
45     }
46
47 }

```

```

<terminated> BinarySearchExample [Java Application] /Applications/Eclipse.app/Contents/Eclipse/plugins/org.eclipse.justi.openjdk.hotspot.jre.full.macosx.aarch64_21.0.9.v20251105-0741/jre/bin/
7
1001 1005 1010 1015 1020 1030 1040
1015
Log Found

```

3. Sort Employee Salaries (Bubble Sort)

```
import java.util.Scanner;
```

```
public class BubbleSortExample{
```

```
    public static void main(String[] args){
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int n = sc.nextInt();
```

```
        int[] salaries = new int[n];
```

```
        for(int i=0; i<n; i++){
```

```
            salaries[i] = sc.nextInt();
```

```
        }
```

```
        for(int i=0; i<n-1; i++){
```

```
            for(int j=0; j<n-1-i; j++){
```

```
                if(salaries[j] > salaries[j+1]){
```

```
                    int temp = salaries[j];
```

```
                    salaries[j] = salaries[j+1];
```

```
                    salaries[j+1] = temp;
```

```
                }
```

```
            }
```

```
        }
```

```
        for(int i=0; i<n; i++){
```

```
            System.out.print(salaries[i] + " ");
```

```
        }
```

```
        sc.close();
```

```
    }
```

```
}
```

```

3 import java.util.Scanner;
4
5 public class BubbleSortExample {
6
7     public static void main(String[] args) {
8         // TODO Auto-generated method stub
9
10        Scanner sc = new Scanner(System.in);
11
12        int n = sc.nextInt();
13
14        int[] salaries = new int[n];
15
16        for (int i = 0; i < n; i++) {
17            salaries[i] = sc.nextInt();
18        }
19
20        // Bubble Sort
21        for (int i = 0; i < n - 1; i++) {
22            for (int j = 0; j < n - 1 - i; j++) {
23                if (salaries[j] > salaries[j + 1]) {
24                    int temp = salaries[j];
25                    salaries[j] = salaries[j + 1];
26                    salaries[j + 1] = temp;
27                }
28            }
29        }
30
31        // Print sorted salaries
32        for (int i = 0; i < n; i++) {
33            System.out.print(salaries[i] + " ");
34        }
35
36        sc.close();
37    }
38 }
39

```

Problems Javadoc Declaration Console X Progress

<terminated> BubbleSortExample [Java Application] /Applications/Eclipse.app/Contents/Eclipse/plugins/org.eclipse.justi.openjdk.hotspot.jre.full.macosx.aarch64_21.0.9.v20251105-074

```

5
45000 32000 55000 28000 60000
28000 32000 45000 55000 60000

```


4. Sort Website Response Time (Insertion Sort)

```
import java.util.Scanner;
```

```
public class InsertionSortExample{
```

```
    public static void main (String[] args){
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int n = sc.nextInt();
```

```
        int[] responseTime = new int[n];
```

```
        for (int i = 0; i < n; i++){
```

```
            responseTime[i] = sc.nextInt();
```

```
        }
```

```
        for (int i = 1; i < n; i++){
```

```
            int key = responseTime[i];
```

```
            int j = i - 1;
```

```
            while (j >= 0 && responseTime[j] > key){
```

```
                responseTime[j+1] = responseTime[j];
```

```
                j--;
```

```
            }
```

```
            responseTime[j+1] = key;
```

```
        }
```

```
        for (int i = 0; i < n; i++){
```

```
            System.out.print(responseTime[i] + " ");
```

```
        }
```

```
        sc.close();
```

```
    }
```

```
}
```

```

4 import java.util.Scanner;
5 public class InsertionSortExample {
6
7     public static void main(String[] args) {
8         // TODO Auto-generated method stub
9         Scanner sc = new Scanner(System.in);
10
11         int n = sc.nextInt();
12         int[] responseTime = new int[n];
13
14         for (int i = 0; i < n; i++) {
15             responseTime[i] = sc.nextInt();
16         }
17
18         // Insertion Sort
19         for (int i = 1; i < n; i++) {
20             int key = responseTime[i];
21             int j = i - 1;
22
23             while (j >= 0 && responseTime[j] > key) {
24                 responseTime[j + 1] = responseTime[j];
25                 j--;
26             }
27
28             responseTime[j + 1] = key;
29         }
30
31         // Print sorted array
32         for (int i = 0; i < n; i++) {
33             System.out.print(responseTime[i] + " ");
34         }
35
36         sc.close();
37     }
38 }
39
40

```

Problems Javadoc Declaration Console X Progress

<terminated> InsertionSortExample [Java Application] /Applications/Eclipse.app/Contents/Eclipse/plugins/org.eclipse.justj.openjdk.hotspot.jre.full.macosx.aarch64_21.0.9.v2

```

5
250 120 320 180 100
100 120 180 250 320

```


5. Prioritize Bug Reports (Selection Sort)

```
import java.util.Scanner;

public class SelectionSortExample {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] priority = new int[n];

        for (int i = 0; i < n; i++) {
            priority[i] = sc.nextInt();
        }

        for (int i = 0; i < n - 1; i++) {
            int minIndex = i;

            for (int j = i + 1; j < n; j++) {
                if (priority[j] < priority[minIndex]) {
                    minIndex = j;
                }
            }

            int temp = priority[i];
            priority[i] = priority[minIndex];
            priority[minIndex] = temp;
        }

        for (int i = 0; i < n; i++) {
            System.out.print(priority[i] + " ");
        }

        sc.close();
    }
}
```

```

3 import java.util.Scanner;
4
5 public class SelectionSortExample {
6
7     public static void main(String[] args) {
8         // TODO Auto-generated method stub
9         Scanner sc = new Scanner(System.in);
10
11         int n = sc.nextInt();
12         int[] priority = new int[n];
13
14         for (int i = 0; i < n; i++) {
15             priority[i] = sc.nextInt();
16         }
17
18         // Selection Sort
19         for (int i = 0; i < n - 1; i++) {
20             int minIndex = i;
21
22             for (int j = i + 1; j < n; j++) {
23                 if (priority[j] < priority[minIndex]) {
24                     minIndex = j;
25                 }
26
27                 // Swap
28                 int temp = priority[i];
29                 priority[i] = priority[minIndex];
30                 priority[minIndex] = temp;
31             }
32
33             // Print sorted array
34             for (int i = 0; i < n; i++) {
35                 System.out.print(priority[i] + " ");
36             }
37
38             sc.close();
39         }
40     }
41 }
42

```

Problems Javadoc Declaration Console X Progress
 <terminated> SelectionSortExample [Java Application] /Applications/Eclipse.app/Contents/Eclipse/plugins/org.eclipse.justj.openjdk.hotspot.jre.full.macosx.aarch64_21.0.9

```

5
9 2 8 1 5
1 2 5 8 9

```